CS182: Introduction to Machine Learning
Lecture 07: Linear Discrimination

Lecture Quiz and Answer Copy

Scope: Linear_Discrimination.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Name: _____ **Student ID:** _____

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Answer sheet — Section A

1	2	3	4	5	6

7	8	9	10	11	12

Answer sheet — Section B

13	14	15	16	17	18

19	20	21	22	23	24

Section A. True/False 12 marks

1. A multiclass argmax classifier with affine scores can have a nonconvex class region even when there are no ties.

2. For any fixed w and w_0, the set $\{x : w^\mathsf{T} x + w_0 = 0\}$ is a separating hyperplane of dimension $d - 1$.

3. Multiplying both the weights and intercept of a binary linear score by a positive constant preserves the predicted labels but generally changes its sigmoid probabilities.

4. The batch algorithm shown in the lecture accumulates errors over all samples before updating the weight vector.

5. For finite binary logits and no numerical rounding, sigmoid produces probabilities strictly between zero and one.

6. For one-hot softmax cross-entropy, a sample generally contributes a gradient to the weights of classes other than its true class.

7. Training K independent one-vs.-rest sigmoid models automatically makes their K outputs sum to one.

8. Adding the same input-dependent affine function to every softmax class score leaves every predicted class probability unchanged.

9. Two learning procedures that both produce a linear class boundary must estimate their parameters by the same statistical objective.

10. When training a one-vs.-one classifier for classes C_i and C_j, examples from other classes are excluded from that particular binary training problem.

11. A standard linear logistic-regression model requires every input feature to have a Gaussian class-conditional distribution.

12. Convexity of unregularized logistic cross-entropy guarantees a finite maximum-likelihood parameter vector on every strictly linearly separable dataset.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

13. Let a batch contain four samples. Compare summed cross-entropy and average cross-entropy at exactly the same weights. Which statements are correct?

(A) At the same learning rate, the average-loss gradient step is one quarter of the summed-loss step.

(B) Choosing the average-loss learning rate four times as large reproduces the summed-loss update.

(C) Zero initialization forces the binary logistic gradient to be zero for every labeled dataset.

(D) Updating weights after every sample is necessarily identical to accumulating the four gradients at fixed weights.

14. Three independently trained affine scores at one test point are $(-2, -1, -3)$. The specified decision rule is argmax over these scores, with no rejection option. Which statements are correct?

(A) Adding ten to every score changes the argmax class.

(B) The point must be rejected because every score is negative.

(C) No label can be assigned until the scores have been individually transformed to probabilities.

(D) The predicted class is C_2.

15. Correct decisions cost zero. Predicting C_1 when the truth is C_2 costs three; the reverse error costs two. At a point with $y = P(C_1 \mid x) = 0.55$, which statements are correct?

(A) The risk of predicting C_2 at the point is 0.9.

(B) The rule must predict C_1 at the given point because $0.55 > 0.5$.

(C) The risk-minimizing rule predicts C_1 when $y > 3/5$.

(D) The equivalent logit threshold is $\log(2/3)$.

16. Consider adding $\lambda\|\theta\|_2^2/2$, with $\lambda > 0$, to a finite-sample linear binary logistic loss, where θ includes the intercept. Which statements are correct?

(A) The added Hessian is λI.

(B) The penalty weight can be selected using validation data.

(C) The penalty necessarily sets some fitted coefficients exactly to zero.

(D) The penalized objective has a unique finite minimizer.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

17. For $\sigma(a) = 1/(1 + e^{-a})$, which statements are correct?

(A) The derivative equals zero at every finite positive logit.

(B) The largest derivative value is $1/4$, attained at $a = 0$.

(C) $\sigma(a + b) = \sigma(a) + \sigma(b)$ for all real a, b.

(D) $\sigma(\log 9) = 0.9$.

18. A binary logistic model predicts $y = 1/4$ for a sample with label $r = 1$ and feature vector $x = (1, -2)^{\mathsf{T}}$. Which statements are correct?

(A) The intercept gradient is $-3/4$.

(B) The negative-gradient update decreases the intercept when the learning rate is positive.

(C) The feature-weight gradient is $(-3/4, -3/2)^{\mathsf{T}}$.

(D) The sample cross-entropy is $\log 4$.

19. Two Gaussian classes have means 1 and -1, shared variance two, and equal priors. Which statements are correct?

(A) This derivation proves that all logistic input distributions must be Gaussian.

(B) The posterior probability of C_1 is $\sigma(x)$.

(C) Changing only the priors, with means and variance fixed, changes only the logit intercept.

(D) Two distinct Gaussian variances always yield the same linear logit.

20. At a point where softmax probabilities are $(0.2, 0.5, 0.3)$, which statements are correct?

(A) $\partial y_2/\partial a_1 = -0.1$.

(B) Every column of the Jacobian sums to one.

(C) $\partial y_2/\partial a_2 = 0.5$.

(D) The Jacobian is diagonal because each numerator uses one score.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

21. Consider $g(x) = x_1^2 + x_2^2 - 1$, with C_1 predicted for $g(x) > 0$. Which statements are correct?

(A) The score is affine in the transformed feature $z = x_1^2 + x_2^2$.

(B) The boundary is a straight line in the original input plane because the coefficients enter linearly.

(C) The boundary is a unit circle in the original input plane.

(D) The model requires x to have a Gaussian distribution.

22. Let $w = (4, -3)^\mathsf{T}$, $w_0 = -5$, and $x = (2, 0)^\mathsf{T}$. Which statements are correct?

(A) The ordinary distance from the origin to the boundary equals minus one.

(B) The Euclidean distance from x to the boundary equals three.

(C) The score at x equals three.

(D) The orthogonal projection is $(38/25, 9/25)^\mathsf{T}$.

23. Three softmax logits are $(0, \log 3, \log 2)$. Which statements are correct?

(A) The second class has probability $1/2$.

(B) Adding seven to every logit preserves all three probabilities.

(C) The predicted class must be rejected because its probability does not strictly exceed $1/2$.

(D) The posterior odds of C_2 against C_3 equal $3/2$.

24. A four-class softmax model has two original input features and an intercept per class. Which statements are correct?

(A) Adding the same three-component augmented weight vector to every class changes the probabilities.

(B) Fixing the reference-class score to zero leaves nine free coefficients.

(C) The symmetric parameterization has twelve coefficients.

(D) For a row-weight matrix $B \in \mathbb{R}^{4\times 3}$, the gradient has the same 4×3 shape.

Section C. Proofs and Formula Derivations

25. Derive the geometry rather than treating the score as a distance **[10 marks]**

(a) Let $g(x) = w^{\mathsf{T}}x + b$, with $w \neq 0$. Prove that w is normal to its zero hyperplane by comparing two points on that hyperplane. **[2 marks]**

(b) Write a point as $x = x_p + rw/\|w\|_2$, where x_p is its orthogonal projection. Derive both the signed-distance formula and an explicit formula for x_p. Explain the corresponding unsigned distance. **[4 marks]**

(c) Prove what happens to the hyperplane, signed distance, and predicted class when (w, b) is multiplied by $c > 0$. Does the corresponding sigmoid probability remain fixed? Include the degenerate case $w = 0$. **[4 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

26. Derive the LDA-to-logistic bridge with its assumptions [10 marks]

(a) Assume $x \mid C_i \sim \mathcal{N}(\mu_i, \Sigma)$ for two classes, with shared positive-definite Σ and positive priors π_i. Derive $\log[P(C_1 \mid x)/P(C_2 \mid x)] = w^\mathsf{T} x + b$, explicitly finding w, b. **[5 marks]**

(b) Solve $a = \log[y/(1-y)]$ for y, and prove the equivalence between the equal-cost posterior threshold and the zero-score threshold. **[2 marks]**

(c) Identify the canceled term that generally remains with unequal class covariances. Explain why the Gaussian derivation does not impose a Gaussian-input requirement on discriminatively fitted logistic regression. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

27. Derive Bernoulli cross-entropy, its gradient, and its limits **[10 marks]**

(a) For conditionally independent labels $r^t \in \{0, 1\}$ with $y^t = \sigma(\theta^\top \widetilde{x}^t)$, derive the conditional likelihood and its negative log-likelihood. Here $\widetilde{x}^t$ contains the intercept feature. **[3 marks]**

(b) Use the chain rule to derive $\nabla_\theta E = \sum_t (y^t - r^t)\widetilde{x}^t$. Then derive the Hessian and prove convexity without treating sigmoid itself as a linear function. **[5 marks]**

(c) For the two observations $(x, r) = (-2, 0), (2, 1)$, set intercept zero and slope $c > 0$. Prove that the unregularized minimum need not be attained at a finite parameter, although the classification error is already zero. **[2 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

28. Derive softmax normalization and the coupled learning rule **[10 marks]**

(a) Take C_K as reference and assume $\log[P(C_i \mid x)/P(C_K \mid x)] = a_i$ for $i < K$. Derive all class probabilities from normalization, then express them in symmetric softmax form. **[3 marks]**

(b) Derive $\partial y_i/\partial a_j$, then derive $\partial \ell/\partial a_j = y_j - r_j$ for one-hot cross-entropy. Give feature-weight and intercept gradients for affine scores. **[4 marks]**

(c) Prove softmax invariance to a common affine score shift, give the free parameter count after fixing a reference class, and explain why a nontrue class still receives a gradient. **[3 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

29. Audit one binary batch update and the resulting decision **[12 marks]**

(a) Train on $(x, r) = (-2, 0), (1, 1), (3, 1)$, starting from slope $w = 0$ and intercept $b = 0$. Use summed cross-entropy and one batch gradient step with learning rate 0.1. Compute the gradients and new parameters. **[4 marks]**

(b) Compute the new probabilities and total cross-entropy. Compare it to the initial loss; exact logarithmic expressions are acceptable. **[4 marks]**

(c) Using the equal-cost threshold, classify all three training points. Does a zero classification error here imply that further loss reduction is impossible? **[2 marks]**

(d) Now suppose predicting C_1 when the truth is C_2 costs four, while the reverse error costs one, with zero correct-decision loss. Derive the posterior threshold and classify the same points under this cost rule. **[2 marks]**

Student response

(a)

(b)

(c)

(d)

Section D. Applications

30. Train a three-class model and check the normalization **[12 marks]**

(a) Consider one training sample $x = (2, -1)^\mathsf{T}$ in class C_3. Use augmented input $\widetilde{x} = (1, 2, -1)^\mathsf{T}$ and a row-weight matrix $B \in \mathbb{R}^{3\times 3}$, initially zero. Compute the one-hot target, probabilities, and loss.**[3 marks]**

(b) Compute the full gradient matrix and update B using learning rate $1/2$. Display every entry and state its dimension. **[5 marks]**

(c) Recompute the logits, the probability of C_3, and the new loss at this sample. Identify the predicted class. **[2 marks]**

(d) Add the row vector $(10, 0, 0)$ to each fitted row of B. Prove whether probabilities change, and describe a numerically stable way to compute the original softmax. **[2 marks]**

Student response

(a)

(b)

(c)

(d)